

LM08 Exchange Rate Calculations

1. Introduction.....	2
2. Cross-Rate Calculations.....	2
3. Forward Rate Calculations.....	3
Summary.....	6

Required disclaimer: IFT is a CFA Institute Prep Provider. Only CFA Institute Prep Providers are permitted to make use of CFA Institute copyrighted materials which are the building blocks of the exam. We are also required to create / use updated materials every year and this is validated by CFA Institute. Our products and services substantially cover the relevant curriculum and exam and this is validated by CFA Institute. In our advertising, any statement about the numbers of questions in our products and services relates to unique, original, proprietary questions. CFA Institute Prep Providers are forbidden from including CFA Institute official mock exam questions or any questions other than the end of reading questions within their products and services.

CFA Institute does not endorse, promote, review or warrant the accuracy or quality of the product and services offered by IFT. CFA Institute®, CFA® and “Chartered Financial Analyst®” are trademarks owned by CFA Institute.

© Copyright CFA Institute

Version 1.0

1. Introduction

This learning module covers:

- How to calculate cross exchange rates
- How to calculate forward exchange rates

2. Cross-Rate Calculations

Given two exchange rates and three currencies, it is possible to determine the third exchange rate. This way of determining the third exchange rate by converting one foreign exchange quote into another is called the cross rate.

Given the two exchange rates below, what is the $\frac{\text{PKR}}{\text{INR}}$ rate?

Currency	Spot rate
PKR/USD	250.0000
INR/USD	80.0000

Solution:

$\frac{\text{PKR}}{\text{INR}} = \frac{\text{PKR}}{\text{USD}} * \frac{\text{USD}}{\text{INR}}$ In this equation, the USD cancels out giving us $\frac{\text{PKR}}{\text{INR}}$.

We are given the value of $\frac{\text{PKR}}{\text{USD}}$. To get the value of $\frac{\text{USD}}{\text{INR}}$, we take the reciprocal of $\frac{\text{INR}}{\text{USD}}$ which is given.

$$\frac{\text{PKR}}{\text{INR}} = 250 * \frac{1}{80} = 3.125$$

Triangular arbitrage: If the implied cross rate is not equal to the quoted cross rate, then an arbitrage opportunity exists and it is called triangular arbitrage. In such cases, one would profit by buying low and selling high. For example, in the case above, if the bank quoted a rate of 3.250 for $\frac{\text{PKR}}{\text{INR}}$, then you may buy INR (sell PKR) for 3.125 and sell INR (buy PKR) for 3.250 to profit from the mispricing.

Example: Cross Rates

(This is based on Example 1 from the curriculum.)

Given the following exchange rates:

Currency	Spot rate
USD/EUR	1.1701
CHF/USD	0.9900
USD/GBP	1.3118

1. Calculate the spot CHF/EUR cross rate.
2. Calculate the spot GBP/EUR cross rate.

Solution to 1:

$$\frac{\text{CHF}}{\text{EUR}} = \frac{\text{USD}}{\text{EUR}} \times \frac{\text{CHF}}{\text{USD}} = 1.1701 \times 0.9900 = 1.1584$$

Solution to 2:

$$\frac{\text{GBP}}{\text{EUR}} = \frac{\text{USD}}{\text{EUR}} \times \left(\frac{\text{USD}}{\text{GBP}} \right)^{-1} = \frac{\text{USD}}{\text{EUR}} \times \frac{\text{GBP}}{\text{USD}} = \frac{1.1701}{1.3118} = 0.8920$$

3. Forward Rate Calculations

Spot Exchange rates: Spot transactions involve the exchange of currencies for immediate delivery. For most currencies, this corresponds to “T+2” delivery, meaning that the exchange of currencies is settled two business days after the trade is agreed to by the two sides of the deal. The exchange rate used for such transactions is called the spot exchange rate.

Forward Exchange rates: Forward contracts are agreements to exchange one currency for another on a future date at an exchange rate agreed upon today. Any exchange transaction that has a settlement date later than T+2 is a forward contract. The exchange rate used for such transactions is called the forward exchange rate.

Forward exchange rates are generally quoted in terms of points or pips. The table below lists the spot rate and several forward rates for $\frac{\text{USD}}{\text{EUR}}$ currency pair. The negative sign indicates that the forward rate is less than the spot rate, i.e., the base currency EUR is weakening relative to the US dollar.

Maturity	Spot rate or forward points
Spot	1.2875
One week	- 0.3
One month	- 1.1
Three months	- 5.5
Six months	- 13.3
Twelve months	- 26.5

To convert forward points into a forward rate, divide the points by 10,000 if the exchange rate uses a four decimal place convention and by 100 if the exchange rate uses a two decimal place convention, and then add to the spot rate.

What is the 12-month forward rate?

$$\text{Forward rate} = 1.2875 + -\frac{26.5}{10,000} = 1.28485$$

At times, forward points can also be expressed as a percentage of the spot rate. For example,

$$-\frac{0.00265}{1.2875} * 100 = -0.21\%$$

Relationship between Spot Rates, Forward Rates, and Interest Rates

The forward exchange rate depends on relative interest rates. To derive the relationship between spot rates, forward rates, and interest rates, assume you have one unit of domestic currency to be invested for let us say, one year. There are two options you may consider:

The exchange rate convention used = $\frac{\text{Foreign currency}}{\text{Domestic currency}} = \frac{\text{Price currency}}{\text{Base currency}}$

Option 1: Invest one unit of domestic currency (base currency) at the domestic risk-free rate i_B for one period.

Amount at the end of the period = $1 + i_B$

Option 2: Convert one unit of domestic currency into foreign currency (base currency) today using the spot rate $\frac{S_P}{B}$. Invest this amount at the foreign risk-free rate i_P for one period.

Determine the forward rate $F_{P/B}$ today at which the price currency will be converted back to base currency.

Amount of units of price currency at the end of the period = $\frac{S_P}{B} (1 + i_P)$

Amount in terms of base (domestic currency) at the end of the period = $\frac{S_P}{B} (1 + i_P) * \frac{1}{\frac{F_P}{B}}$

Instructor's Note: By using forward rate, any foreign exchange risk was eliminated in this transaction.

Since the risk of these two investments is the same, they should generate equivalent returns. As an investor, you should be indifferent between the two as it is a no-arbitrage relationship.

$$1 + i_B = \frac{S_P}{B} (1 + i_P) * \frac{1}{\frac{F_P}{B}}$$

The above equation can be rewritten as:

$$F_{\frac{P}{B}} = S_{\frac{P}{B}} * \frac{1 + i_P}{1 + i_B}$$

where:

$F_{\frac{P}{B}}$ = forward rate

i_P = risk – free rate of the price currency

i_B = risk – free rate of the base currency

Let us use some values now to compute the forward rate. The spot rate for INR/USD $S_{\frac{INR}{USD}} = 100$.

The risk-free rate for INR is 10% and the risk-free rate for dollar (base currency) is 1%.

$$\text{Forward rate } F_{\frac{INR}{USD}} = S_{\frac{INR}{USD}} * \frac{1 + i_{INR}}{1 + i_{USD}} = 100 * \frac{1 + 0.1}{1 + 0.01} = 108.9108$$

Some important points to be noted:

- The currency with the higher (lower) interest rate will always trade at a discount (premium) in the forward market.
- In our example above, USD was the base currency with a risk-free rate of 1%, while INR was the price currency with a risk-free rate of 10%. The forward rate calculated was 108.91 which means that the currency with the lower interest rate (USD) appreciated, while the currency with the higher interest rate (INR) depreciated. (Trading at a discount means the currency depreciates. Trading at a premium means the currency appreciates.)
- This relationship ensures that there is no arbitrage. The only forward rate that prevents arbitrage is 108.91. Otherwise, for any rate greater/lesser than 108.91 traders can exploit the mispricing by buying low and selling high.
- If the forward contract is for x days, make an adjustment based on the x/360 convention unless told otherwise. Refer to the equation below.

$$F_{\frac{P}{B}} = S_{\frac{P}{B}} * \frac{1 + (i_P * x/360)}{1 + (i_B * \frac{x}{360})}$$

Example

Given the following data, calculate the 37-day forward rate for JPY/CAD:

Canadian dollar risk-free interest rate = 3.97%

Japanese yen risk-free interest rate = 0.23%

Spot rate $S_{\frac{\text{JPY}}{\text{CAD}}} = 100.87$

Solution:

$$\text{Forward rate} = 100.87 \left(\frac{1 + 0.0023 * \frac{37}{360}}{1 + 0.0397 * \frac{37}{360}} \right) = 100.87 * (1.000236/1.00408) = 100.48 \text{ JPY/CAD}$$

Summary

LO: Calculate and interpret currency cross-rates.

The computations for cross rates are a simple case of algebraic manipulation wherein the quotes should be set up in a way that the common currency cancels out.

LO: Explain the arbitrage relationship between spot and forward exchange rates and interest rates, calculate a forward rate using points or in percentage terms, and interpret a forward discount or premium.

Arbitrage relationship between spot and forward exchange rates and interest rates:

- $F_{P/B} = S_{P/B} (1 + i_p) / (1 + i_B)$
- In this formula the interest rates have been de-annualized, if annualized rates are given, de-annualize using $n/360$.

Calculating a forward rate using points or in percentage terms:

- If spot rate is 1.4000 USD/EUR and the forward quote is spot plus 200 points, the forward becomes 1.4200.
- To convert points to decimal divide by 10,000: $200 / 10,000 = 0.02$.
- The point is the last digit of the spot rate quote.
- We can also express the forward rate as a percentage above or below the spot rate.

Interpreting a forward discount or premium:

- If forward rate > spot rate, the base currency is said to be trading at a forward premium.
- If forward rate < spot rate, the base currency is said to be trading at a forward discount.
- The currency with the higher (lower) interest rate will always trade at a discount (premium) in the forward market.
- Interest rate differential = $i_p - i_B$
- Percentage change in spot rates is proportional to the interest rate differential.